Smart Farm Management System

Relational Tables - Sample Data

FARM

Farm ID	Name	Area
F001	Green Valley	150 acres
F002	Sunny Meadows	120 acres
F003	Blue Ridge	200 acres
F004	Golden Field	175 acres
F005	Silver Farm	130 acres

CROP

Crop ID	Crop Name	Crop Type
C001	Wheat	Cereal
C002	Tomato	Vegetable
C003	Orange	Fruit
C004	Corn	Cereal
C005	Apple	Fruit

FARMERS' COMMUNITY

Name	Member Count	Establishment Date
Agri Group	50	2000-01-01
Green Farmers	80	2005-06-15
Organic Society	30	2010-11-20
Rural Union	100	1995-08-10
Crop Experts	60	2015-03-05

LOCATION

Farm ID	Farm Location
F001	East Valley
F002	West Hills
F003	North Plains
F004	South Ridge
F005	Central Basin

FARMER

ID	Fname	Minit	Lname	Bdate	Address	Phone Number	Farm ID	Name FC
101	John	L	Doe	1980-05-14	Riyadh	0501234567	F001	Agri Group
102	Jane	N	Smith	1992-08-22	Jeddah	0509876543	F002	Green Farmers
103	Paul	F	Taylor	1975-11-03	Dammam	0504321987	F003	Organic Society
104	Emily	M	Clark	1985-01-15	Mecca	0503456789	F004	Rural Union
105	Michael	K	Johnson	1990-06-25	Khobar	0507654321	F005	Crop Experts

WORKS_WITH

Farmer ID	Supplier ID
F001	SP001
F002	SP002
F003	SP003
F004	SP004
F005	SP005
F001	SP004
F003	SP004
F003	SP001
F005	SP001
F002	SP004

PLANTED_IN

Field ID	Supplier ID
FL001	C001
FL002	C002
FL003	C003
FL004	C004
FL005	C005
FL004	C002
FL001	C004
FL001	C004
FL003	C002
FL003	C002

SUPPLIER

ID	Fname	Minit	Lname	Phone Number	Email	Address	Product Type
SP001	Jane	A	Smith	555-6789	b2@fake.net	123 Farm Lane	Fertilizers
SP002	Tom	B	Brown	555-4321	a1@test.com	456 Green St	Pesticides
SP003	Lucy	C	Davis	555-4321	c3@demo.org	789 Sunny Rd	Seeds
SP004	Mark	D	Wilson	555-2109	e5@mail.co	321 Blue Ave	Machinery
SP005	Anna	E	Johnson	555-6543	d4@temp.io	654 Golden Blvd	Greenhouse Supplies

FIELD

Field ID	Area	Soil Type	Farm ID	Sensor ID	Irrigation ID
1	50	Loamy Soil	Loamy	505	1001
2	35	Sandy Soil	Sandy	502	1002
3	60	Clay Soil	Clay	503	1003
4	45	Paety Soil	Silty	501	1004
5	80	Silty Soil	Peaty	504	1005

SENSOR

Sensor ID	Sensor Type	Sensor Location
501	Temperature	Field 1
502	Humidity	Field 2
503	Soil Moisture	Field 3
504	Temperature	Field 4
505	Soil Moisture	Field 5

SMART IRRIGATION

Irrigation ID	Irrigation Type	Field ID
I001	Drip	FL001
I002	Sprinkler	FL002
I003	Flood	FL003
I004	Manual	FL004
I005	Pivot	FL005

WEATHER

Weather ID	Temperature	Date	Humidity	Weather condition	Farm ID
W001	28°C	2024-03-15	65%	Sunny	F001
W002	30°C	2024-04-01	70%	Cloudy	F002
W003	25°C	2024-02-10	80%	Rainy	F003
W004	22°C	2024-05-05	60%	Windy	F004
W005	35°C	2024-01-20	50%	Hot	F005

CROP RECORD

Crop ID	Record ID	Planting Date	Harvest Date	Environmental Conditions	Quantity Harvested
C001	R001	2024-03-15	2024-07-10	Sunny with moderate rainfall	500
C002	R002	2024-04-01	2024-08-20	High humidity, occasional storms	320
C003	R003	2024-02-10	2024-06-30	Dry with low rainfall	450
C004	R004	2024-05-05	2024-09-15	Wet season with consistent rainfall	600
C005	R005	2024-01-20	2024-05-25	Cold temperatures, low humidity	250